GOV-AR: Risk-Bounded Multi-Tenant LLM Admission and Routing under Delayed Token-Cost Feedback

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Abstract—Shared enterprise LLM gateways face a control problem that post-hoc reporting does not solve: output-token cost is unknown at admission time, several requests may already be in flight, and provider telemetry may settle late. We study this problem through GOV-AR, a governance-aware admission, routing, and reservation path for multi-tenant AI gateways under delayed token-cost feedback. The current artifact combines hard governance filtering, tenant-isolated settled-versus-reserved budget accounting, request-level reservation and idempotent settlement, and first-class `admit`, `queue`, `reject`, and `abstain` outcomes. The implementation spans a reproducible research workspace, an operator-side `gov-ar-admission` service, an optional PostgreSQL-backed ledger, and a proxy-triggered live path. We report trace-driven evidence across delayed settlement, multi-tenant contention, drift, fault handling, scalability, live Azure validation, and ablation. The current results establish a reproducible systems-and-method artifact, but not yet a full prompt-complete or submission-complete Q1 study; we document the remaining gaps explicitly.

Index Terms—LLM gateways, admission control, delayed feedback, Kubernetes operators, multi-tenant systems, FinOps, sovereign AI, Azure OpenAI.

1 Introduction

Cost control for enterprise LLM gateways is difficult because the financial commitment of a request is partly hidden until the response completes. A budget controller that only tracks already settled usage ignores costs already engaged by in-flight requests. At the opposite extreme, reserving for worst-case output length is safe but wasteful. GOV-AR targets this middle ground: it decides whether to admit, queue, reject, or abstain, selects a deployment, and commits a reservation while respecting hard governance constraints.

The operator used in this work already contains telemetry ingestion, cost attribution, budget tracking, sovereignty enforcement, route recommendation, and Kubernetes-native policy surfaces. The missing piece is a reserve-dispatch- settle control path for uncertain and delayed-cost requests. This article uses the operator as the implementation substrate for that scientific decision problem rather than as the article's end goal.

2 Motivation and Failure Example

Consider two tenants sharing the same premium deployment. Tenant A has already spent most of its budget window. Tenant B has a fresh budget, but several long responses are still in flight. If the controller admits a new request based only on settled spend, it may over-commit. If it reserves at maximum output length for every request, it may reject requests that would have remained safe under calibrated reservations. This motivates a controller with explicit liability tracking and conservative fallback when uncertainty grows.

3 Related Work

Adaptive LLM routing under budget constraints has been studied in recent work such as Adaptive LLM Routing under Budget Constraints [1]. Cost-aware cascades and routers such as FrugalGPT [2], RouteLLM [3], HYBRID LLM [4], and RouteLMT [5] show that query-dependent selection can improve quality-cost trade-offs. Conformal routing adds distribution-free safety at the routing level [6]. Recent ACL 2026 work extends this space toward broad model pools, length-aware routing, and multi-turn trajectories [7], [8], [9]. From the systems side, output-length-aware serving and unknown-length-aware scheduling motivate reservation and length prediction mechanisms [10], [11], [12]. Foundational theory for delayed feedback and resource-constrained online decisions appears in online learning under delayed feedback [13] and contextual bandits with knapsack-style constraints [14], [15].

The remaining gap is their combination with hard governance filtering, tenant-isolated budget liability, explicit queue and abstain outcomes, and a Kubernetes-native implementation path. The literature review is materially improved, but still not saturated to the final target size and stopping rule required by the original prompt.

4 Problem Formulation

For each request r from tenant t, GOV-AR chooses one of `admit(model,reservation)`, `queue`, `reject`, or `abstain`. Before scoring, the candidate set is hard-filtered by allowed provider, allowed zone, sensitivity compatibility, routability, approval state, and model availability. The cost model is

$$C_{t,m} = p_{in}(m) \cdot n_{in}(t) + p_{out}(m) \cdot N_{out}(t, m),$$

where $N_{out}(t, m)$ is unknown at admission time.

The core accounting abstraction is

$$\mathrm{available}_t(k) = \mathrm{budget}_t - \mathrm{settled}_t(k) - \mathrm{reserved}_t(k).$$

This separates already observed spend from in-flight financial liability.

5 GOV-AR DESIGN

The current design includes four coupled components.

5.0.0.1 Governance filter.: Non-compliant models are eliminated before any soft ranking step.

5.0.0.2 Reservation policy.: The present artifact supports empirical quantile reservation and mean-plus-standard-deviation reservation over output-length estimates.

5.0.0.3 Ledger and settlement.: The request ledger stores reservation amount, tenant, selected deployment, pricing version, reservation mode, settlement status, and cancellation state. Settlement is idempotent for repeated delivery of the same event identifier.

5.0.0.4 Conservative fallback.: When drift or insufficient evidence invalidates a confident decision, the controller can queue or abstain instead of silently continuing with stale assumptions.

6 SAFETY AND RISK ANALYSIS

The strongest safety property currently evidenced is deterministic accounting discipline for the implemented path: reservations are unique per request, duplicate settlement identifiers are tolerated idempotently, conflicting duplicate settlements are rejected, and cancellation releases reserved liability. The current artifact does not yet provide a full prompt-grade probabilistic guarantee with calibration intervals, coverage analysis by tenant, and a final risk-bounded theorem. Scientific claims should therefore be read as observed behavior plus partial design-level safety, not as a complete proven guarantee.

7 KUBERNETES AND GATEWAY IMPLEMENTATION

The implementation is split between the research workspace in `article3/` and operator-side code under `operateur/`. The research workspace provides the admission kernel, ledger, predictors, replay harness, experiments, provenance, and manuscript assets. The operator-side slice adds:

- `internal/govar/` for CRD-to-snapshot translation and decision backends;
- `cmd/gov-ar-admission` for `/v1/admit`, `/v1/settle`, `/v1/cancel`, and `/v1/liability/{tenant}`;
- an optional PostgreSQL-backed transactional ledger;
- chart templates to deploy the admission service;
- a proxy-triggered path in `header-proxy`;
- webhook-sidecar propagation of GOV-AR configuration through pod annotations.

This is a meaningful operator integration slice, but it is still not a final Envoy-native `ext_proc` or external-authorization production path.

8 EXPERIMENTAL METHODOLOGY

The artifact contains a frozen protocol file, raw outputs, processed outputs, and reproducible orchestration scripts. However, the current execution depth must be distinguished from the ideal prompt protocol. The evidence package is reproducible, but most experiment families remain below the requested scale in seed count, request count, scenario breadth, and statistical completion. The current repository now also includes an automatically generated statistical summary over the available E1 and E2 campaign outputs, using paired bootstrap intervals over run-level metric differences.

9 BUDGET SAFETY AND UTILIZATION

In the deterministic E1 summary, quantile reservation admitted two requests and settled a total cost of 12.0 with 2.0 slack and zero overshoot. The mean-plus-standard-deviation variant admitted one request, queued one, settled 7.0, and also had zero overshoot. In the campaign comparison, the trade-off shifts: one variant can reduce overshoot while increasing slack. This supports the central practical tension between utilization and conservatism. In the current paired campaign analysis, the E1 quantile variant shows higher overshoot than mean-plus-standard-deviation, with a mean overshoot difference of 1.18 and a paired bootstrap 95% interval of approximately $[0.19, 2.35]$, while also using less slack by about 1.6 units.

10 MULTI-TENANT ROUTING

In the deterministic E2 trace, both compared variants admitted three requests and queued one, but the quantile variant incurred 0.5 overshoot while the mean-plus-standard-deviation variant used more slack and avoided overshoot. Tenant-level settled totals were recorded separately, confirming that the current harness tracks tenant-isolated accounting. That said, the prompt-grade multi-tenant study with six rich tenant profiles and large-scale scenarios is still incomplete. In the available E2 campaign analysis, the sign of the trade off reverses: quantile reservation reduces overshoot relative to mean-plus-standard-deviation by about 1.62 units on average, with a paired bootstrap 95% interval of approximately $[-2.60, -0.59]$.

11 DRIFT AND FAULT INJECTION

The drift experiment demonstrates an observable conservative mode. After a measured ratio drift above threshold, the artifact falls back to a safer candidate and abstains when no sufficiently supported candidate remains. Fault injection validates duplicate-settlement handling, expiry release, and telemetry-insufficiency abstention. What is still missing is the full 20-scenario fault matrix with the repetition depth requested in the prompt.

12 Scalability and Live Azure Validation

The local scalability sweep replays synthetic traces up to 1000 requests and measures replay throughput in the lightweight harness. These numbers should be read only as local replay-engine observations. The Azure live validation successfully discovered and invoked three real deployments spanning Azure OpenAI and Foundry-served Mistral, with HTTP 200 responses and traceable token usage or latency metadata. This is a real live validation, but still not the full prompt-scale experiment of at least 300 successful requests per model across multiple independent windows.

13 Ablation and Sensitivity

The ablation package compares reservation and queue-policy variants and shows that aggressive under-reservation increases overshoot while disabling queueing turns deferred decisions into direct refusals. This is useful mechanistic evidence, but it still falls short of the full A0–A9 by scenario matrix requested in the prompt.

14 Threats to Validity

The main threats are scale, dataset realism, and maturity bias. Synthetic traces do not fully capture production prompt distributions. Replay throughput does not equal end-to-end gateway throughput. The literature review is stronger than the initial state but not yet fully saturated. The manuscript has moved closer to a scientific structure, yet it is still not a final Q1 submission.

15 Discussion

The strongest contribution of the current repository state is not a polished paper claim, but a hardening of the implementation substrate: operator audit, reserve-settle scaffold, operator-side admission service, optional PostgreSQL ledger, local Kind automation, live Azure validation, and a now much clearer prompt-compliance audit trail. The main remaining work is therefore not finding small wording tweaks, but closing the genuine scientific and experimental gaps.

16 Conclusion

GOV-AR frames enterprise LLM routing as a joint admission, reservation, and settlement problem under delayed token-cost feedback and hard governance constraints. The present artifact already demonstrates a reproducible decision kernel, a Kubernetes-oriented implementation path, local Docker/Kind/Helm execution, and real Azure validation against existing deployments. It does not yet satisfy the full original prompt, and it is not yet a finished Q1 article. The next mandatory steps are full prompt-scale experimentation, stronger statistical analysis, deeper Envoy-native integration, complete evidence tracking, and final publication packaging.

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